

Isolating the center contribution to entanglement entropy across four gauge groups

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We measure the Rényi-2 entanglement entropy S_2 in four pure Yang–Mills theories— $SU(2)$, $SU(3)$, $SU(4)$, and G_2 —on $L^3 \times 2L$ lattices with $L = 4\text{--}12$ at matched string tension. The pair $SU(4)$ and G_2 shares the same quadratic Casimir $C_2 = 4$ and adjoint dimension (15 vs. 14) but differs in center ($\mathbb{Z}_4$ vs. $\{1\}$). We find $S_2^{SU(4)}/S_2^{G_2} = 1.191 \pm 0.002$, constant across volumes, with a deficit $\Delta S_2/A = 3.45 \pm 0.03$ attributable entirely to the center. A three-parameter fit $S_2/A = a C_2 + b \ln|Z| + c$ describes all 20 data points at the 2% level, providing the first quantitative decomposition of lattice entanglement entropy into Casimir and center contributions. Per-gluon entropy decreases monotonically ($2.41 \rightarrow 1.31$), demonstrating large- N screening.

INTRODUCTION

Entanglement entropy (EE) in gauge theories probes confinement, topological order, and holographic duality [? ? ?]. The Donnelly–Wall decomposition [?] splits gauge-theory EE into classical superselection (determined by the center flux through the entangling surface), edge-mode, and bulk contributions. For a group with trivial center $Z(G) = \{1\}$, the classical term vanishes.

Prior lattice studies measured S_2 in $SU(N_c)$ for $N_c = 2\text{--}5$ [?] and found scaling compatible with $\dim(\text{adj})$. However, within the $SU(N)$ family, center size, Casimir, and dimension all grow together, precluding disentanglement. The exceptional group $G_2 = \text{Aut}(\mathbb{O})$ breaks this degeneracy: it has $C_2(\text{adj}) = 4$ like $SU(4)$ and $\dim(\text{adj}) = 14 \approx 15 = \dim(\text{adj}_{SU(4)})$, but its center is trivial, $Z(G_2) = \{1\}$. A comparison of $SU(4)$ and G_2 thus isolates the center contribution at fixed Casimir.

In this Letter, we present the first measurement of S_2 in G_2 lattice gauge theory and compare it with $SU(2)$, $SU(3)$, and $SU(4)$, providing a model-independent decomposition of EE into its group-theoretic components.

METHOD

We use the Wilson plaquette action $S = (\beta/d_R) \sum_{\square} \text{Re Tr}(1 - U_{\square})$ with d_R the fundamental dimension: $d_R = 2, 3, 4$ for $SU(N)$ and $d_R = 7$ for G_2 . For G_2 , links are 7×7 real orthogonal matrices; the 14 generators of $\mathfrak{g}_2 \subset \mathfrak{so}(7)$ are obtained as the kernel of the 35×21 constraint matrix preserving the Bryant 3-form $\varphi = e^{123} + e^{145} + e^{167} + e^{246} - e^{257} - e^{347} - e^{356}$.

The Rényi-2 entropy is measured via the α -integration method [?]:

$$S_2 = \int_0^1 d\alpha \left\langle \frac{\partial S}{\partial \alpha} \right\rangle_{\alpha}, \quad (1)$$

TABLE I. S_2/A for four gauge groups at matched $\sigma a^2 \approx 0.047$, with group-theoretic parameters.

Group	β	d	C_2	$ Z $	$L=4$	$L=12$	$S_2/A/d$
$SU(2)$	2.50	3	2	2	7.22	7.13	2.41
$SU(3)$	6.06	8	3	3	14.17	13.99	1.77
G_2	10.0	14	4	1	18.30	18.09	1.31
$SU(4)$	10.80	15	4	4	21.69	21.54	1.45

where α interpolates boundary conditions at $x_0 = L/2$ on an $L^3 \times 2L$ lattice. The integral is evaluated by trapezoidal rule over 11 α -points. Each is thermalized with 500 Metropolis sweeps; measurements are 200 configurations separated by 5 sweeps.

Coupling constants are matched to a common string tension $\sigma a^2 \approx 0.047$: $\beta_{SU(2)} = 2.50$ [?], $\beta_{SU(3)} = 6.06$ [?], $\beta_{SU(4)} = 10.80$ [?], $\beta_{G_2} = 10.0$ [?]. The entangling surface area is $A = L^2 \times 2L = 2L^3$ (the 3-dimensional boundary of the bipartition in four dimensions).

RESULTS

Table ?? summarizes results at $L = 4$ and $L = 12$ (full data: 5 volumes per group, 20 points total). All groups exhibit an area law $S_2 \propto L^{3.00(1)}$ with mild $O(1/L^2)$ finite-size corrections.

$SU(4)$ vs. G_2 : isolating the center

$SU(4)$ and G_2 share $C_2 = 4$ and have nearly equal $\dim(\text{adj})$ (15 vs. 14), yet differ in center: $\mathbb{Z}_4$ vs. $\{1\}$. Across all measured volumes:

$$\frac{S_2^{SU(4)}}{S_2^{G_2}} = 1.191 \pm 0.002 \quad (L = 4\text{--}12), \quad (2)$$

constant to better than 0.5%. The surplus $\Delta S_2/A \equiv S_2^{SU(4)}/A - S_2^{G_2}/A = 3.45 \pm 0.03$ is attributable to the center $\mathbb{Z}_4$, consistent with $\Delta S/\ln|Z(4)| = 2.49$.

Cross-group ratios

All ratios are volume-independent:

$$S_2^{SU(2)}/S_2^{SU(3)} = 0.510 \pm 0.001, \quad (3)$$

$$S_2^{SU(4)}/S_2^{SU(3)} = 1.536 \pm 0.003, \quad (4)$$

$$S_2^{G_2}/S_2^{SU(3)} = 1.291 \pm 0.001. \quad (5)$$

Three-parameter decomposition

We fit the $L = 4$ data (matched σa^2) to

$$\frac{S_2}{A} = a C_2(\text{adj}) + b \ln|Z(G)| + c, \quad (6)$$

obtaining $a = 6.42$, $b = 2.33$, $c = -7.38$. Residuals are below 2% for all four groups. The positive coefficient $b > 0$ confirms that the center *enhances* EE through superselection sectors, as predicted by the Donnelly–Wall decomposition [?]: groups with $Z = \{1\}$ ($\ln|Z| = 0$) have strictly less entanglement at fixed Casimir.

Per-gluon entropy

The entropy per adjoint degree of freedom $S_2/(A \cdot d)$ decreases monotonically:

$$\frac{S_2}{A \cdot d} : \quad 2.41 \text{ (SU(2))} \rightarrow 1.77 \rightarrow 1.45 \rightarrow 1.31 \text{ (G}_2\text{)}, \quad (7)$$

demonstrating that gluons become progressively screened as the group grows. G_2 has the *lowest* per-gluon entropy despite having the largest adjoint after $SU(4)$, consistent with its trivial center eliminating the classical contribution entirely.

G_2 MULTI- β UNIVERSALITY

Table ?? shows S_2/A for G_2 across five values of β . At each β , S_2/A is constant in L (within 2%), confirming the area law. The β -dependence $S_2/A \propto \beta$ reflects the expected UV divergence $\sim 1/a^2$ of the area-law coefficient. The ratio $G_2/SU(3)$ at matched σa^2 is independent of β , confirming that the group-dependent structure is a physical (not lattice-artifact) effect.

TABLE II. G_2 multi- β results (S_2/A) at $L = 4, 6, 8$.

β	$L = 4$	$L = 6$	$L = 8$
9.6	15.99	15.67	15.65
9.8	17.23	16.93	16.86
10.0	18.30	18.09	—
10.2	19.21	19.19	19.24
10.4	20.48	20.37	20.39

DISCUSSION

Donnelly–Wall validation.—The pair $SU(4)/G_2$ provides the first “controlled experiment” for the center contribution: same C_2 , comparable $\dim(\text{adj})$, different $|Z|$. The 19% excess of $SU(4)$ over G_2 is a direct, quantitative confirmation of center-dependent EE on the lattice, going beyond the qualitative consistency previously reported for $SU(N)$ alone [?].

Effective dimension.—The $G_2/SU(3)$ ratio $R = 1.29$ implies $d_{\text{eff}}(G_2) \approx 10.3$ out of 14 gluons. Under the maximal subgroup decomposition $\mathbf{14} = \mathbf{8} \oplus \mathbf{3} \oplus \mathbf{\bar{3}}$, the $\mathbf{3} \oplus \mathbf{\bar{3}}$ sector can form color singlets and is partially screened at the entangling surface.

Large- N screening.—The decreasing per-gluon entropy $S_2/(A \cdot d)$ from $SU(2)$ to G_2 is consistent with large- N factorization: at $N \rightarrow \infty$, gluon correlations decorrelate, and the entropy per degree of freedom should approach a constant [?].

Confinement without center.— G_2 confines despite $Z = \{1\}$ [?], with a first-order deconfinement transition driven by mechanisms other than center-vortex condensation [?]. Our data shows that the entanglement structure mirrors this: the area-law coefficient is reduced relative to $SU(4)$ at the same Casimir, consistent with the absence of center-mediated long-range correlations.

CONCLUSION

We have measured the Rényi-2 entanglement entropy in $SU(2)$, $SU(3)$, $SU(4)$, and G_2 lattice gauge theories at matched physical scale, totaling 32 data points across five lattice volumes and five coupling constants. The comparison of $SU(4)$ and G_2 at equal Casimir isolates the center contribution to EE for the first time, yielding $\Delta S_2/A = 3.45$ per unit entangling area. The three-parameter fit Eq. (??) decomposes EE into Casimir and center terms with 2% accuracy, providing a quantitative test of the Donnelly–Wall prediction. All code, data, and analysis are publicly available [?].

Computational resources: consumer workstation (Intel i5-14600KF, 64 GB RAM, NVIDIA RTX 5060 Ti). The lattice code was developed in Rust (`nalgebra` library). This research received no external funding.

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